
netcheck Documentation

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Gene C

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WIREGUARD TOOL MANUAL:

1	netcheck	1
1.1	Overview	1
1.2	Key features	1
1.3	Recent Changes	1
1.4	Getting Started	1
1.5	Options	2
2	Appendix	5
2.1	Installation	5
2.2	Dependencies	5
2.3	License	5

NETCHECK

1.1 Overview

netcheck : Check network connectivity

1.2 Key features

- netcheck uses ping to check network status of multiple hosts/ips
- Checking a single remote site is insufficient to distinguish whether the path to that remote is the only route unavailable, or if all routes are down. We therefore recommend checking 2 hosts simultaneously. While more may be better, we have found using 2 reliable remote hosts eliminates all false negatives.
- Data is saved to file(s)
- netcheck-plot can be used to plot the saved data.

1.3 Recent Changes

1.11.0

- Use meson / meson-python for build / package management,
- periodic code review

1.4 Getting Started

netcheck is designed to be run periodically, saving results to files from which netcheck-plot can be used to create plots showing packet loss over time.

By default netcheck is silent unless there is packet loss. This is convenient when run from cron.

Using the `-v` option allows it to show output even when no packet loss is seen.

For example you may want to run this command from cron, perhaps every 15 minutes:

```
netcheck -d /var/lib/netcheck 1.1.1.1 8.8.8.8
```

it will report nothing unless both remote hosts have some packet loss. Since occasional packet drops are not uncommon, its good to have 2 or more hosts for the check. You can use `-any` option to have it report if any host has some packet loss.

If no hosts are provided, then default hosts are `1.1.1.1` and `8.8.8.8`.

Using the `-d <dir>` option has packet loss data saved into files under the specified directory. With the above command observed packet losses will be saved to 4 files:

```
ncheck-1.1.1.1
ncheck-8.8.8.8
ncheck-merged.any
ncheck-merged.all
```

One file for each host and the `ncheck-merged.any` records packet seen by any host and `ncheck-merged.all` has the data when all hosts had packet loss seen at the same time.

The `-I`, `-Interface` option can be used to specify which network interface the check is to be run from. Not to be confused with the `-i`, `-interval` which specifies the interval used between each ping packet.

To generate a plot for any of the files simply run:

```
netcheck-plot -f /opt/netcheck/ncheck-merged.all
```

which by default will create the pdf file

```
./pdf/ncheck-merged.all.pdf
```

For list of options please use `-help` for either application.

1.4.1 How It Works

Netcheck runs ping in parallel on each host listed using asyncio. By default 100 packets are sent to each host, though this can be changed using the `-num` option.

1.5 Options

Available options from `-help`:

netcheck:

```
positional arguments:
  hosts                host(s) comma separated if more than one (None)

options:
  -h, --help            show this help message and exit
  -a, --any              Report error if any host has loss instead of all
  ->hosts (False)
  -b, --base BASE       base name for output (ncheck)
  -d, --dir DIR         directory to save output ( )
  -i, --interval INTERVAL ping interval (0.25)
  -I, --Interface INTERFACE ping interface or IP address to use as source ( )
  -4, --ip4-only        Use IPv4 only (False)
  -6, --ip6-only        Use IPv6 only (False)
  -n, --num NUM         num pings (100)
  -np, --no_parallel    No parallel (False)
  -t, --test            test mode - generate fake lost packets(False)
  -v, --verb            verbose mode (False)
```

netcheck-plot:

options:

-h, --help	show this help message and exit
-d, --dir DIR	Directory to write plot pdf file (./pdf)
-e, --end END	End time : absolute or relative days (all)
-f, --file FILE	Input csv file (data)
-o, --out OUT	Output PDF : if not specified will be based on 
 input filename	
-s, --start START	Start time : absolute or relative days (all)
-sh, --show	Show plot in terminal (False)

APPENDIX

2.1 Installation

Available on * [Github](#) * [Archlinux AUR](#)

On Arch you can build using the provided PKGBUILD in the packaging directory or from the AUR. All git tags are signed with arch@sapience.com key which is available via WKD or download from <https://www.sapience.com/tech>. Add the key to your package builder gpg keyring. The key is included in the Arch package and the source= line with *?signed* at the end can be used to verify the git tag. You can also manually verify the signature:

```
git tag -v <tag-name>
```

To build manually, clone the repo and

```
./scripts/do-build  
./scripts/do-install <destination-directory>
```

2.2 Dependencies

Run Time :

- python (3.13 or later)
- python-pandas
- python-matplotlib
- pyconcurrent

Building Package :

- git
- meson
- meson-python
- rsync

2.3 License

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